



1.5 Complex numbers (HL)

Number and algebra · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A complex number $z=a+bi$ has real part a and imaginary part b , where $i^2=-1$. In polar/Euler form $z=r(\cos\theta+i\sin\theta)=r\cdot e^{i\theta}$, where $r=|z|=\sqrt{a^2+b^2}$ is the modulus and $\theta=\arg(z)$ is the argument. Multiplication/division are easiest in polar form (moduli multiply/divide, arguments add/subtract). De Moivre's theorem states $z^n = r^n(\cos n\theta + i\sin n\theta)$, used for powers and finding n th roots.

Worked example: follow the reasoning

Find $(1+i)^4$ using De Moivre's theorem.

$1+i$ has $r=\sqrt{2}$, $\theta=\pi/4$.

$$(1+i)^4 = (\sqrt{2})^4(\cos(4\cdot\pi/4)+i\sin(4\cdot\pi/4)) = 4(\cos\pi + i\sin\pi) = 4(-1+0i) = -4.$$

Quick check · multiple choice

What is the modulus of $z = 3-4i$?

- A 1
- B 5
- C 7
- D 25

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Let $z_1 = 2+2i$.

[8]

- (a) Write z_1 in the form $r(\cos\theta+i\sin\theta)$, giving exact values of r and θ .
- (b) Hence find z_1^6 in Cartesian form $a+bi$.
- (c) Find the two square roots of -4 in the form $a+bi$.

Final answer: _____



Answers and explanation

Question 1 [8]

- (a) $r = \sqrt{4+4} = 2\sqrt{2}$, $\theta = \arctan(2/2) = \pi/4$. So $z_1 = 2\sqrt{2}(\cos(\pi/4) + i\sin(\pi/4))$.
(b) $z_1^6 = (2\sqrt{2})^6(\cos(6\pi/4) + i\sin(6\pi/4)) = 512(\cos(3\pi/2) + i\sin(3\pi/2)) = 512(0-i) = -512i$.
(c) $-4 = 4(\cos\pi + i\sin\pi)$. Square roots: $\sqrt{4}(\cos(\pi/2 + k\pi) + i\sin(\pi/2 + k\pi))$ for $k=0,1 \rightarrow$
 $2(\cos(\pi/2) + i\sin(\pi/2)) = 2i$, and $2(\cos(3\pi/2) + i\sin(3\pi/2)) = -2i$.

Quick check answer

B. $|z| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.